

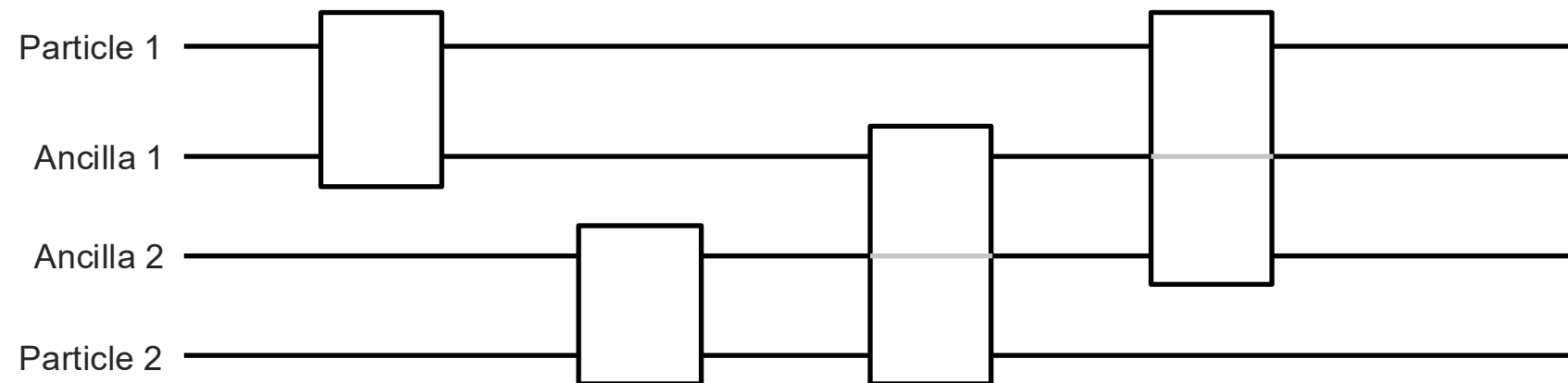
Can Gravity be a Pairwise Local Classical Channel?

Jamie Breault

Jan 4, 2018

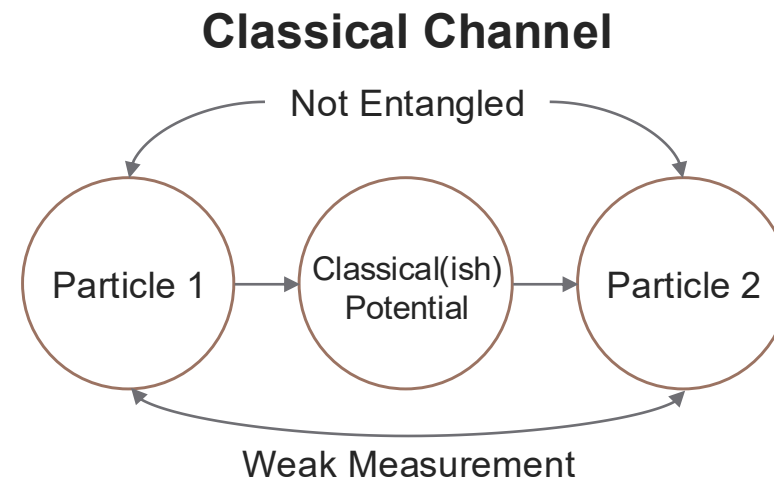
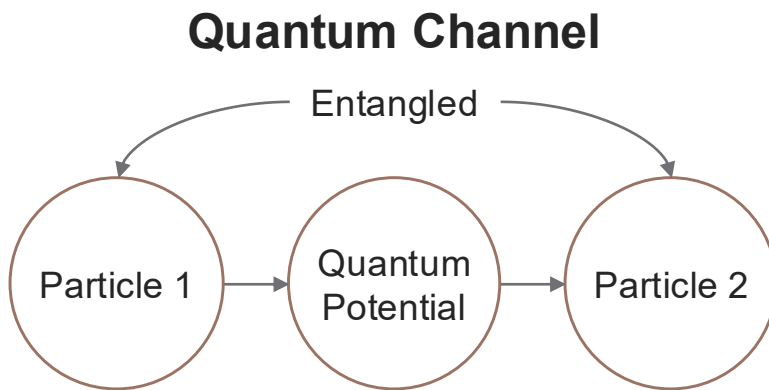
Pairwise

- Particles interact gravitationally via ancillae
- Ancillae are force carrying particles
- There are 2 ancillae for each pair of particles



Classical Channel

- Does not increase entanglement
- Equivalent to ancillae conducting weak measurements and feedback



Comparing Theory to Experiment

- Atomic fountain experiments
 - Prepare atoms in superpositions
 - Determine visibility as a function of superposition size
 - Visibility is how long the atom can stay in a superposition
- We can obtain maximum theoretical visibility as a function of experiment parameters
 - Decoherence rate minimization
 - Heating rate minimization

Decoherence Rate Minimization

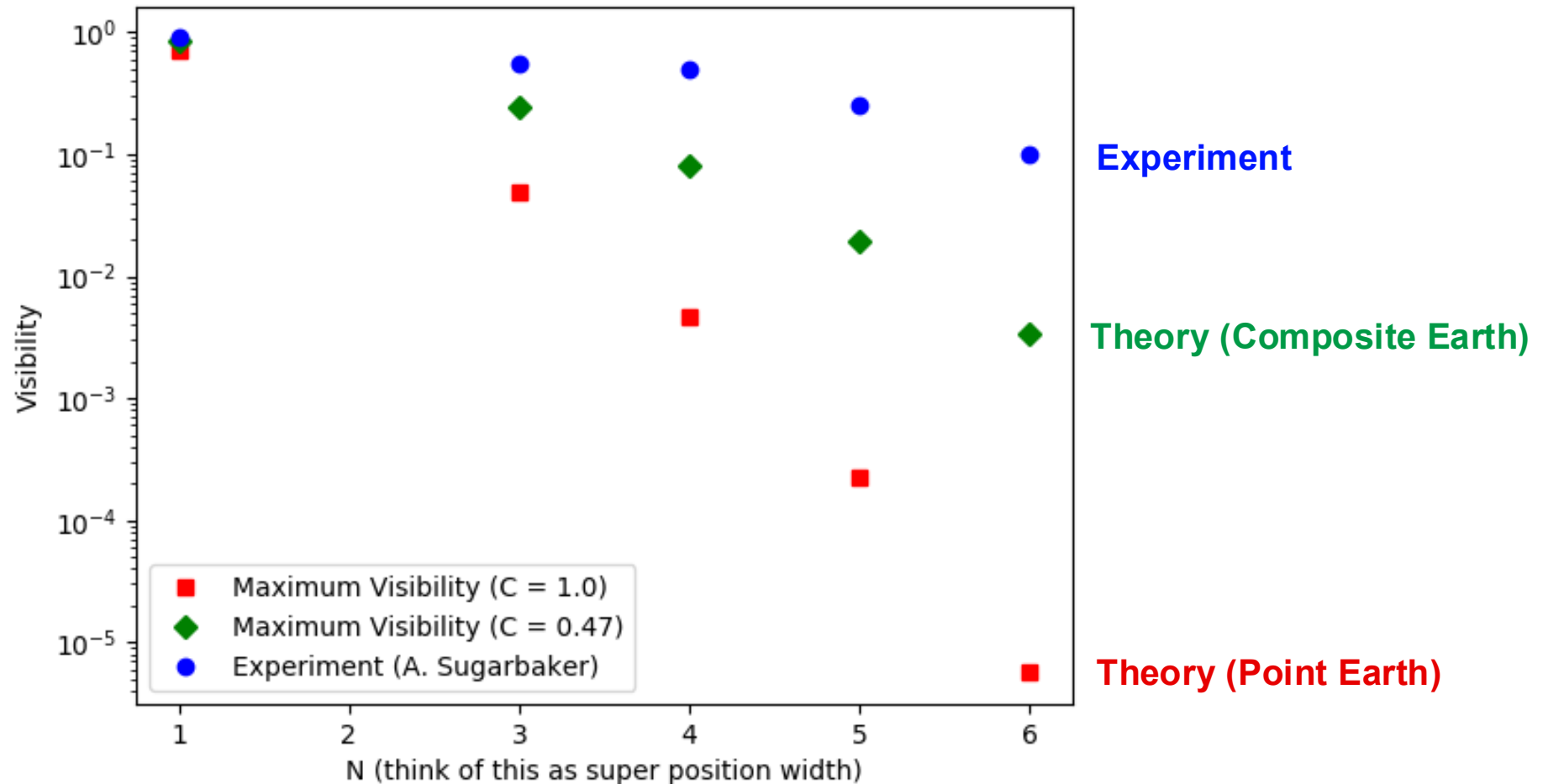
Master Equation	$\dot{\rho} = -\frac{i}{\hbar}[\hat{H}_0 + \hat{V}_0, \rho] - \left(\frac{1}{4D} + \frac{K^2 D}{4\hbar^2}\right) \sum_{i=1}^2 [\hat{x}_i, [\hat{x}_i, \rho]]$
Decoherence Rate	$\Gamma_{KTM} = \left(\frac{1}{4D} + \frac{K^2 D}{4\hbar^2}\right) \Delta x^2$
Minimum Decoherence Rate (2 particle case)	$\Gamma_{KTM}^{min} = \frac{K}{2\hbar} \Delta x^2$
Minimum Decoherence Rate (multi-particle case)	$\Gamma_{KTM}^{min} = C \frac{K}{2\hbar} \Delta x^2$ C = 1.0 (Point Earth) C = 0.47 (Composite Earth)
Maximum Visibility	$V^{max} = e^{-2 \int_0^T \Gamma_{KTM}^{min} dt} = e^{-C \frac{8G\hbar k^2 M_{\oplus} T^3 N^2}{3m_A R_{\oplus}^3}}$

K → Measurement Strength

D → Feedback Strength

Δx → Superposition Size

Comparing to Experiment



Heating Rate Minimization

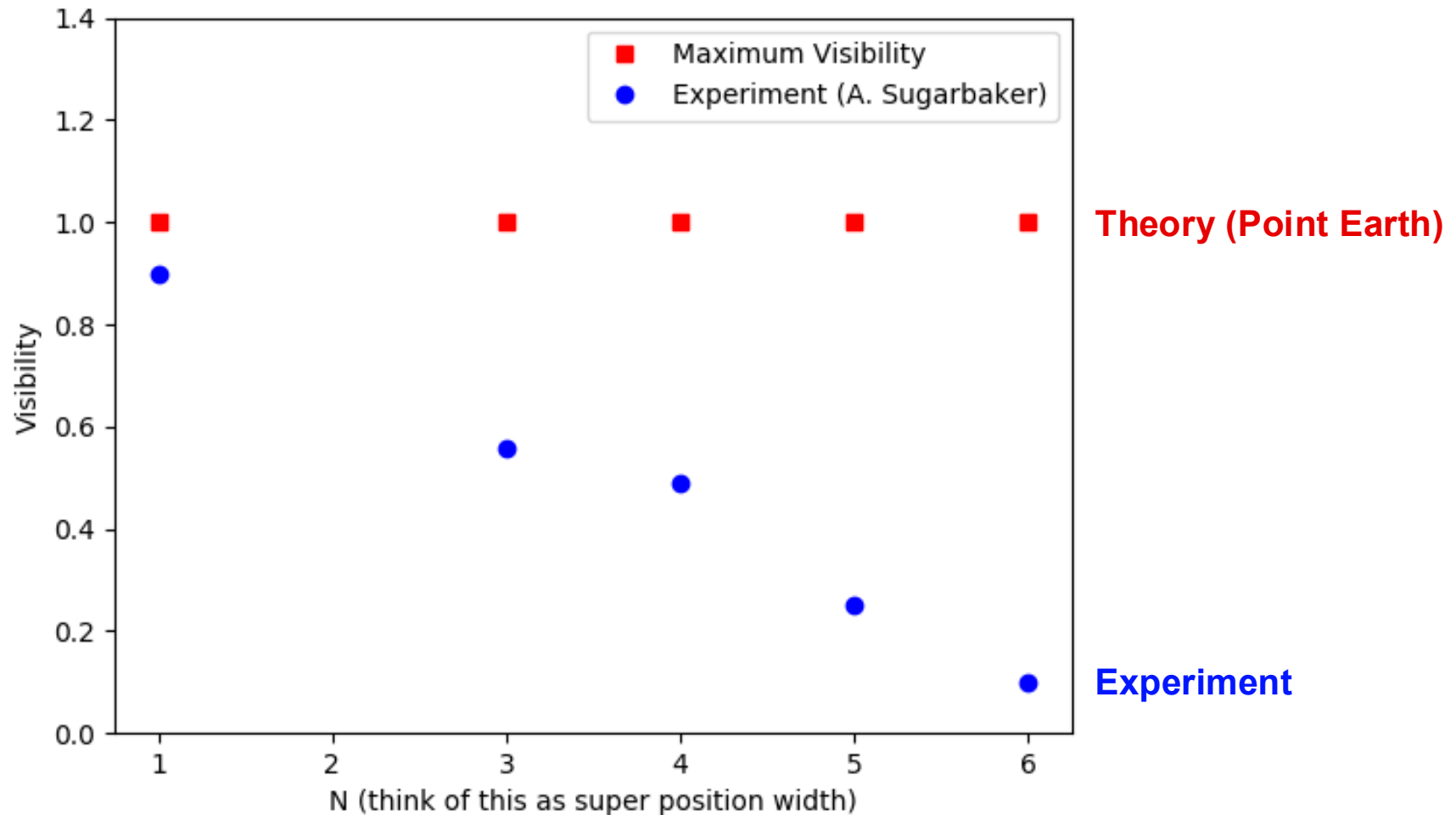
Master Equation	$\dot{\rho} = -\frac{i}{\hbar}[\hat{H}_0 + V_0, \rho] - \left(\frac{1}{4D_1} + \frac{K^2 D_2}{4\hbar^2}\right)[\hat{x}_1, [\hat{x}_1, \rho]] - \left(\frac{1}{4D_2} + \frac{K^2 D_1}{4\hbar^2}\right)[\hat{x}_2, [\hat{x}_2, \rho]]$
Heating Rate	$R_h = \langle \dot{H}_0 \rangle = \text{Tr}[H_0 \dot{\rho}] = \frac{\hbar^2}{m_1} \left(\frac{1}{4D_1} + \frac{K^2 D_2}{4\hbar^2}\right) + \frac{\hbar^2}{m_2} \left(\frac{1}{4D_2} + \frac{K^2 D_1}{4\hbar^2}\right)$
D_1 and D_2 at Minimum Heating Rate	$D_1 = \frac{\hbar}{K} \sqrt{\frac{m_2}{m_1}} \quad D_2 = \frac{\hbar}{K} \sqrt{\frac{m_1}{m_2}}$
Master Equation at Minimum Heating Rate	$\dot{\rho} = -\frac{i}{\hbar}[\hat{H}_0 + V_0, \rho] - \frac{K}{2\hbar} \left(\sqrt{\frac{m_1}{m_2}} [\hat{x}_1, [\hat{x}_1, \rho]] + \sqrt{\frac{m_2}{m_1}} [\hat{x}_2, [\hat{x}_2, \rho]] \right)$
Equation of Motion of the Atom	$\dot{\rho}_A = -\frac{i}{\hbar}[\hat{H}_A + V, \rho_A] - \frac{K}{2\hbar} \sqrt{\frac{m_A}{M_{\oplus}}} [\hat{x}_A, [\hat{x}_A, \rho]]$
Maximum Visibility	$V_{heat}^{max} = (V_{deco}^{max}) \sqrt{\frac{m_A}{M_{\oplus}}}$

$K \rightarrow$ Measurement Strength

$D \rightarrow$ Feedback Strength

$\Delta x \rightarrow$ Superposition Size

Comparing to Experiment



Concluding Remarks

- Only considering decoherence rate, gravity is not classical under this model
- Considering heating rate too, gravity may be classical
- Next step is the global case (1 ancilla per particle)

Acknowledgements

- Natacha Altamirano
- Robert Mann

Classical Channel Check

- Entanglement never increases if the matrix $Y - 2ig\sigma$ has no negative eigenvalues where

$$\dot{\rho} = -\frac{i}{\hbar}[\hat{H}_0, \rho] - ig[\hat{x}_1\hat{x}_2, \rho] - \frac{1}{4} \sum_{k=1}^2 Y_{ij}[\hat{x}_i, [\hat{x}_j, \rho]] \quad \sigma = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\dot{\rho} = -\frac{i}{\hbar}[\hat{H}_0, \rho] - i\frac{K}{\hbar}[\hat{x}_1\hat{x}_2, \rho] - \frac{K}{2\hbar} \left(\sqrt{\frac{m_1}{m_2}}[\hat{x}_1, [\hat{x}_1, \rho]] + \sqrt{\frac{m_2}{m_1}}[\hat{x}_2, [\hat{x}_2, \rho]] \right)$$

$$g = \frac{K}{\hbar} \quad Y = \frac{2K}{\hbar} \begin{pmatrix} \sqrt{\frac{m_1}{m_2}} & 0 \\ 0 & \sqrt{\frac{m_2}{m_1}} \end{pmatrix} \quad Y - 2ig\sigma = \frac{2K}{\hbar} \begin{pmatrix} \sqrt{\frac{m_1}{m_2}} & -i \\ i & \sqrt{\frac{m_2}{m_1}} \end{pmatrix}$$

- $Y - 2ig\sigma$ has eigenvalues of 0 and $\frac{2K(m_1+m_2)}{\hbar\sqrt{m_1m_2}}$